

CMBSG Hardware Implementation and Development Process Guide

VOLUME 1

Procurement, Vendor Specifications & Bill of Materials

REFERENCE DESIGN BASIS: 10,000 Nm³/HR CMBSG MODULE
DOCUMENT CLASS: ENGINEERING DELIVERABLE — PRE-CONSTRUCTION

Read Before Use

This volume corrects a set of process-basis errors identified during independent verification of the CMBSG technical reference (mass balance, Hatta number, pressure drop, and material property inconsistencies). Every equipment specification in this volume is built on the **corrected** basis shown in Section 1. Do not procure equipment against figures from the original research paper without cross-checking Section 1 first. Where a specification depends on chemistry that is genuinely hazardous (HF, NaBH₃CN/HCN precursor handling), this volume flags it explicitly and defers final process safety sign-off to a qualified Process Safety Management (PSM) review — this document is an engineering planning and procurement aid, not a substitute for a stamped, jurisdiction-approved engineering package.

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1 How to Use This Manual Set

1.1 Document Set Overview

This is Volume 1 of an eight-volume hardware implementation package for the Continuous Multiphase Biomineralization and Solidification Grid (CMBSG). The full set is structured to mirror how a real capital-equipment project is actually documented and executed, not how a research paper is written:

Volume	Title	Primary Audience
1	Procurement, Vendor Specs & BOM (<i>this volume</i>)	Procurement, project engineering
2	Mechanical Installation & Assembly Manual	Site construction, mechanical contractors
3	Electrical, Instrumentation & Control Systems	E&I contractors, controls engineers
4	Commissioning, Startup & Performance Testing	Commissioning team, plant management
5	Safety Systems, HAZOP Close-out & PSM	HSE, process safety engineers
6	Standard Operating Procedures	Plant operators, shift supervisors
7	Maintenance Manual & Spare Parts	Maintenance planners, technicians
8	Operator Training & Competency Framework	Training coordinators, HR

1.2 How This Volume Is Organized

Section 1 establishes the corrected engineering basis of design (this is the single most important section in the volume — every quantity used later traces back to it). Sections 2–6 give vendor-ready equipment specifications stage by stage. Section 7 consolidates everything into a master Bill of Materials. Section 8 covers procurement scheduling and long-lead items. Section 9 defines vendor qualification criteria. Section 10 covers site civil/structural prerequisites. The appendices provide blank datasheet and Inspection & Test Plan (ITP) templates for use with your own vendor RFQs.

1.3 Notation and Tagging Convention

Equipment tags follow the convention established in the original technical reference: a letter code (R=reactor, T=tank, P=pump, HP=hydraulic press, PM=pug-mill, HC=hydrocyclone, CT=curing tunnel, MX=mixer) followed by a stage-prefixed number (1xx=Stage 1, 2xx=Stage 2, 3xx=Stage 3, 4xx=Stage 4). This is preserved unchanged from the reference document so that this implementation package cross-references cleanly against it.

2 Corrected Engineering Basis of Design

2.1 Why This Section Exists

An independent verification pass on the CMBSG technical reference paper found a set of numerical errors, the most consequential being that the total flue-gas mass flow (12,900 kg/hr) had been used interchangeably with the CO₂-specific mass flow throughout the mass-balance and revenue sections. Before specifying a single pump, vessel, or instrument range, this section fixes that basis so that every number in Sections 2–10 is internally consistent.

Summary of Corrections Applied

- **CO₂ mass flow:** corrected from 12,900 kg/hr to **2,749 kg/hr** ($10,000 \text{ Nm}^3/\text{hr} \times 14 \text{ vol\%} \div 22.414 \text{ Nm}^3/\text{kmol} \times 44.01 \text{ kg/kmol}$). The 12,900 kg/hr figure is retained *only* where it correctly describes total flue-gas mass flow (Stage 1 thermal/fan duty).
- **Annual CO₂ captured:** corrected from three mutually-inconsistent figures (91,912 t/yr; 1,856 t/yr; and an unlabeled intermediate) to a single reconciled **19,586 t/yr** at 95% capture and 7,500 hr/yr.
- **Fly-ash design load:** the original document used 45 kg/hr in the input specification table but 450 kg/hr in the Stage 1 energy balance with no reconciliation. This volume adopts **45 kg/hr as the nominal mass-balance / Stage 4 solids-loading value**, and retains **450 kg/hr strictly as the conservative thermal-duty design case** for Stage 1 equipment margin (i.e., the quench and ID fan are sized for the higher figure; the press/curing solids throughput is sized for the nominal figure).
- **Stage 4 solids throughput and block production:** recomputed bottom-up from the corrected CaCO₃ yield. Result: **2,010 blocks/hr**, which is coincidentally very close to the original document's stated 2,035 blocks/hr (see Section 1.4) — meaning **Stage 4 press and curing tunnel sizing in the original reference is approximately valid and is retained**, while the CO₂-tonnage and CCTS-revenue figures that fed off the wrong intermediate number are not, and are corrected here.
- **Stage 3 mesh tower duty:** re-derived from the corrected CO₂ capture load (2,611.5 kg/hr captured, not 12,255 kg/hr). This affects enzyme (CA-KR1) dosing mass balance and the absorption column's assumed CO₂ loading, both corrected in Section 4.
- **Hatta number / mass-transfer enhancement (Stage 3):** independent recalculation gives $Ha = 0.088$ (not 0.877) and enhancement factor $E_{en} \approx 1.0$ (not 1.25) from the stated inputs. This does not change equipment *sizing* directly, but it materially changes the expected CO₂ capture efficiency the tower can deliver at the stated enzyme loading — flagged here and carried into Volume 4 (Performance Testing) as an open item requiring pilot verification rather than assumed design performance.
- **Titanium material properties:** Appendix A.4 of the original reference lists density, sound speed, and acoustic impedance for Ti Grade 2 that are mutually inconsistent ($4,510 \text{ kg/m}^3 \times 3,700 \text{ m/s} \neq 27.3 \times 10^6$), and the Stage 3 ultrasonic amplitude calculation used $2,710 \text{ kg/m}^3$ (aluminum's density) mislabeled as titanium. This volume uses the **correct handbook values for Ti Grade 2** (density $4,510 \text{ kg/m}^3$, longitudinal sound speed $6,070 \text{ m/s}$) throughout Section 4's ultrasonic transducer specification.
- **Stage 1 throat pressure drop:** corrected from a stated 1.47 kPa to the value the formula and inputs actually produce, **0.147 kPa**; the SIS high- ΔP trigger threshold (Volume 5) is unaffected since 5 kPa remains a valid, conservative alarm setpoint either way.

2.2 Reference Plant Capacity

Table 1: Corrected reference input stream specification (design basis for all equipment in this volume).

Parameter	Value	Basis / Note
Volumetric flow rate, Q_g	10,000 Nm ³ /hr	Unchanged from reference
Inlet temperature, T_{in}	180 °C	Unchanged
Total flue gas mass flow	12,900 kg/hr	Correct as TOTAL gas flow (1.29 kg/Nm ³)
CO ₂ concentration	14.0 vol%	Unchanged
CO₂ mass flow (corrected)	2,749 kg/hr	$10,000 \times 0.14 / 22.414 \times 44.01$
SO ₂ concentration	1,200 mg/Nm ³	Unchanged; SO ₂ chain was already correct
Fly ash load (nominal, mass balance)	45.0 kg/hr	Adopted per input spec table
Fly ash load (Stage 1 thermal design case)	450 kg/hr	Conservative margin, quench/fan sizing only
Operating hours	7,500 hr/yr	Unchanged
Slag feed rate	250 kg/hr	Unchanged (independent of CO ₂ error)
Chitosan feed rate	18.32 kg/hr	Unchanged

2.3 Corrected Carbon and Sulfur Mass Balance

Table 2: Corrected top-level mass balance (per hour, dry basis, 95% CO₂ / 98% SO₂ capture).

Stream	CO ₂ [kg/hr]	SO ₂ [kg/hr]	Solids [kg/hr]
Raw flue gas (in)	2,749	12.0	45.0 (ash, nominal)
Slag leachate (in)	—	—	120 (slag, Ca-bearing fraction)
Chitosan matrix (in)	—	—	18.32 (polymer)
Cleaned gas (out, 5%/2% slip)	137.4	0.24	< 0.5 (PM)
CaCO ₃ produced	2,611.5 captured		5,933.8 (as CaCO ₃)
Gypsum (CaSO ₄ ·2H ₂ O) produced	11.76 captured		31.6
Total solids to Stage 4 press			6,028.7

2.4 Reconciliation: Why Stage 4 Equipment Sizing Survives the Correction

Engineering Note

The corrected total solids feed to Stage 4 (6,028.7 kg/hr) is within 1.2% of the original reference's stated 6,105 kg/hr. At 3.0 kg/block, this gives **2,009.6 blocks/hr**, essentially matching the original 2,035 blocks/hr claim. **This means the Stage 4 press, curing tunnel, and block-handling equipment specified in the original reference did not need to be re-sized** — the error was isolated to how CO₂ tonnage was reported and monetized (Sections 13–14 of the reference, i.e., CCTS credits and "cost per tCO₂ avoided"), not to the physical solids-handling equipment train. Volume 1 therefore keeps Stage 4 mechanical specifications close to the original design, while Section 4 of this volume (Stage 3) is revised more substantially because enzyme dosing and column CO₂ loading *do* depend directly on the corrected CO₂ mass flow.

3 Master Equipment List

Table 3: Master Equipment List (MEL) — all major tagged equipment, corrected basis.

Tag	Description	Duty / Capacity	Primary Material
<i>Stage 1 — Gas Conditioning</i>			
D-101	Inlet ductwork	10,000 Nm ³ /hr, Ø600mm	Carbon steel, Al ₂ O ₃ -tile lined
V-101	Venturi quench throat	Ø280mm, 45 m/s throat velocity	95% Al ₂ O ₃ ceramic liner
T-101	Quench sump tank	3,500 L, 45 min retention	Super Duplex UNS S32750
F-101	Induced draft (ID) fan	10,000 Nm ³ /hr @ 5 kPa	FRP / 316L wetted parts
P-101	Sump recirculation pump	15 m ³ /hr @ 2 bar (sized to sump retention duty)	Super Duplex
<i>Stage 2 — Matrix Blending</i>			
R-201	Slag leaching reactor	15,000 L, Ø2400mm	Glass-lined carbon steel
T-202	Dissolution tank	12,000 L working	316L SS mirror-polished
MX-203	Helical static mixer	1,800mm × Ø150mm	Schedule 80 CPVC
P-202	Recirculation pump	30 m ³ /hr @ 3 bar	Hastelloy C-276
<i>Stage 3 — Kinetic Mesh Absorber</i>			
C-301	Mesh absorber tower	2.5×2.5m, 12m height	Titanium Grade 2 (mesh), FRP shell
US-301/2/3	Ultrasonic transducer sets	40 kHz, corrected amplitude spec	Langevin-type, Ti Grade 2 horn
DS-301	CA-KR1 enzyme dosing skid	corrected dosing rate, Section 4	316L SS
<i>Stage 4 — Consolidation Press</i>			
PM-401	Twin-shaft pug-mill	1.5 m ³ , 45 RPM	Hardox 400 blades
HC-401	Hydrocyclone	75mm apex, 1.5 bar ΔP	Polyurethane-lined
HP-401	4-cavity hydraulic press	250 t, 200 bar, 30s dwell	Hardox 400 dies
CT-401	Curing tunnel	18m, 55–60 °C, 90% RH	Insulated steel, PVC liner

4 Stage 1 Equipment Specifications — Gas Conditioning & Venturi Quench

4.1 Design Duty Summary

Stage 1 Process Duty

- Gas flow: 10,000 Nm³/hr, 180→45 °C, saturated outlet
- Heat duty: 536 kW (verified correct, Section 4.2 of technical reference)
- Evaporative water demand: 806 kg/hr
- Design particulate load: 450 kg/hr (thermal case) / 45 kg/hr (nominal, downstream solids)
- Target PM₁₀ removal: verification found the stated 32 μm cut-diameter is inconsistent with the paper's own Calvert-model inputs (correct value ≈0.15 μm). **Action:** procure the venturi to the vendor's own certified performance curve for the stated throat velocity and L/G ratio, do not specify cut-diameter as a design input — verify PM₁₀ removal empirically during commissioning (Volume 4).

4.2 Venturi Quench Stack (V-101) — Vendor Specification Sheet

Table 4: V-101 Venturi Quench — vendor RFQ datasheet.

Parameter	Specification
Service	Flue gas quench and primary particulate scrubbing
Inlet gas flow	10,000 Nm ³ /hr ±2%, 180 °C, +200 Pa gauge
Outlet gas condition	45 °C saturated, ≤0.7% CO ₂ slip target downstream
Throat diameter	280 mm (fixed geometry; vendor may propose adjustable throat if quoted with a positioner)
Throat velocity	45 m/s design point
Inlet duct diameter	600 mm
Throat liner material	95% Al ₂ O ₃ ceramic, 50 mm thickness, minimum 10-year erosion life at stated ash loading
Body material	Carbon steel shell, rubber-lined or ceramic-tile lined per vendor standard
L/G ratio (design)	1.2 L/m ³
Pressure drop (corrected)	0.15 kPa nominal at design point; vendor to certify against own performance curve, SIS trip set at 5 kPa (see Volume 5)
Connections	600 mm inlet flange (gas), 150 mm quench water inlet, bottom flanged outlet to sump T-101
Design code	ASME Section VIII Div. 1 (pressure boundary), or equivalent local pressure vessel code
Inspection requirement	100% radiography of shell longitudinal welds; liner adhesion test per vendor's standard QC

4.3 Quench Sump Tank (T-101) — Vendor Specification Sheet

Table 5: T-101 Quench Sump — vendor RFQ datasheet.

Parameter	Specification
Service	Acidic condensate collection, ash slurry concentration (40 wt% target)
Working volume	3,500 L, sized for 45 min retention at design flow
Diameter	1,800 mm
Design pH range	2.5–3.5 continuous, transient excursion to pH 1.5
Material of construction	Super Duplex stainless steel UNS S32750 (PREN ≥ 40)
Design chloride limit	≤ 200 ppm (verify against actual makeup water quality before final MOC sign-off)
Level instrumentation	Radar or guided-wave level transmitter, 0–100% range, alarm at 20%/85%
Nozzles	Top: venturi outlet inlet (flanged); side: overflow to Stage 4 bypass line; bottom: agitator/drain; instrument nozzles per P&ID (Volume 3)
Design code	ASME Section VIII Div. 1 or API 650 (atmospheric) depending on final pressure rating chosen by vendor
Corrosion allowance	1.5 mm minimum over 15-year design life

4.4 Induced Draft Fan (F-101) and Ductwork

Table 6: F-101 ID Fan — vendor RFQ datasheet.

Parameter	Specification
Service	Induced draft, downstream of Stage 1 quench (handles saturated 45 °C gas)
Flow rate	11,648 Nm ³ /hr actual (corrected for 45 °C, see Volume 1 §6 tower-sizing cross-check)
Static pressure rise	Sized for total system ΔP : Stage 1 (0.15 kPa) + Stage 3 tower (max 1.8 kPa per SIS-C) + ductwork losses; specify minimum 3.5 kPa fan rise with 25% margin
Motor power	35 kW nominal (per original reference electrical load summary)
Wetted parts material	FRP or 316L SS coating, rated for saturated acidic gas (pH 3.5 condensate carryover)
Speed control	VFD, 40–100% turndown
Noise limit	≤ 85 dBA at 1m per OSHA/Factory Act requirement

5 Stage 2 Equipment Specifications — Matrix Blending Loop

5.1 Design Duty Summary

Stage 2 Process Duty

- Slag feed: 250 kg/hr BOF slag (unaffected by CO₂ correction)
- Chitosan feed: 18.32 kg/hr, dissolved to 3.0 wt% solution
- Target matrix output pH: 4.0–4.5 (protonation >99.7% per Henderson-Hasselbalch, verified correct)
- **Correction applied:** the original saturation index calculation (claimed -0.58) was independently recomputed as $+0.42$ from the paper's own stated concentrations — a positive SI means the design's intended chemistry runs closer to supersaturation than assumed. **Action:** specify T-202 with an additional acid-dosing trim loop (see below) to actively hold SI <0 rather than relying on the nominal Ca²⁺ /chitosan ratio alone, and require the vendor's control system to log real-time pH and conductivity as a proxy SI indicator.

5.2 Leaching Reactor (R-201) — Vendor Specification Sheet

Table 7: R-201 Slag Leaching Reactor — vendor RFQ datasheet.

Parameter	Specification
Service	Acid-mediated BOF slag Ca ²⁺ leaching, pH 2.5–3.5
Volume	15,000 L, Ø2,400 mm
Retention time	Minimum 12 min (verified: Shrinking Core Model gives 692 s = 11.5 min for 90% conversion at $k_{\text{diff}} = 8 \times 10^{-4} \text{ s}^{-1}$); specify 20 min nominal for operating margin
Construction	Glass-lined carbon steel, jacketed
Jacket service	Cooling water, 25–30 °C supply (leaching is mildly exothermic, $\Delta G_{298}^{\circ} = -193 \text{ kJ/mol}$ for the CaO+2H ⁺ reaction)
Agitation	Top-entry, glass-lined impeller, variable speed 20–80 RPM
Slag feed	Vibratory feeder, 250 kg/hr, Ø800 mm hopper
Acid dosing	Dilute HNO ₃ or acetic acid (CH ₃ COOH preferred to avoid nitrate contamination of cured block per original reference rationale)
Instrumentation	pH (redundant pair), temperature, level, conductivity
Design code	ASME Section VIII Div. 1, glass-lining per DIN 28136 or equivalent

5.3 Dissolution Tank (T-202) with SI Trim Control

Table 8: T-202 Dissolution Tank — vendor RFQ datasheet (revised with SI trim loop).

Parameter	Specification
Service	Chitosan/ Ca^{2+} /CA-KR1 matrix blending prior to Stage 3
Working volume	12,000 L
Construction	316L SS, mirror-polished interior, baffled
NEW: SI trim sub-loop	Add a dedicated acid micro-dosing loop (metering pump, 0–5 L/hr, pH-cascade controlled) targeting a continuous SI < −0.3 safety margin, not just the nominal −0.58 design point that verification could not reproduce.
Instrumentation	pH, conductivity, temperature, level; SI soft-sensor computed in DCS from real-time Ca^{2+} (ion-selective electrode) and pH
Mixing	Baffled, low-shear agitation compatible with Carreau–Yasuda pseudoplastic rheology of the chitosan matrix ($\mu_0 = 1.5 \text{ Pa}\cdot\text{s}$ zero-shear)
Design code	ASME Section VIII Div. 1

5.4 Static Mixer, Pumps, and Piping

Table 9: Stage 2 rotating equipment and piping — vendor RFQ datasheet.

Parameter	Specification
MX-203 static mixer	Schedule 80 CPVC, 1,800 mm length, $\varnothing 150 \text{ mm}$, helical element count per vendor sizing for the stated non-Newtonian rheology
P-202 recirculation pump	Centrifugal, Hastelloy C-276 wetted parts, $30 \text{ m}^3/\text{hr}$ @ 3 bar (verified: Hagen-Poiseuille pressure drop calc gives 0.115 bar over 25m of DN80 CPVC pipe — pump head has ample margin)
Transfer piping	Schedule 80 CPVC, DN80, flanged joints (chitosan solution is aggressive to many elastomers/metals at pH 4.0–4.5; CPVC rated to 93°C is standard, see Volume 2 for installation details)
CA-KR1 dosing skid	See Section 4.4 (moved to Stage 3 boundary since enzyme is dosed at the Stage 2/3 interface per original P&ID)

6 Stage 3 Equipment Specifications — Kinetic Mesh Absorber Column

6.1 Corrected Design Duty Summary

Recalculated Stage 3 Duty

- **CO₂ to be captured: 2,611.5 kg/hr (95% of the corrected 2,749 kg/hr feed)** — not 12,255 kg/hr as in the original reference. This is a $\sim 4.7\times$ reduction in the absorption duty the column must actually deliver.
- **Practical implication:** the mesh tower geometry (2.5m $\times$ 2.5m $\times$ 12m, 27s gas residence time) was sized on gas *volumetric* residence time, which is unaffected by the CO₂ mass-flow error, so **tower dimensions do not need to change**. What changes is the enzyme dosing rate and the expected mass of CaCO₃ actually produced per hour — both revised below.
- **Hatta number finding carried forward:** independent recalculation gives $Ha=0.088$ (paper claimed 0.877, a $10\times$ decimal error) and enhancement factor $E_{en}\approx 1.0$ (paper claimed 1.25). **This is the single highest-priority open item in the entire hardware package.** If $E_{en}\approx 1.0$ is correct, the CA-KR1 enzyme is not providing the mass-transfer enhancement the 95–99% capture claim depends on, and **pilot-scale verification of actual capture efficiency at the stated enzyme loading is mandatory before this column is procured at full scale** (see Volume 4, Section 2, Performance Test Protocol).

6.2 Mesh Absorber Tower (C-301) — Vendor Specification Sheet

Table 10: C-301 Kinetic Mesh Absorber — vendor RFQ datasheet.

Parameter	Specification
Service	CO ₂ /SO ₂ enzymatic + chemical absorption, three-tier gradient mesh
Shell dimensions	2.5 m $\times$ 2.5 m square cross-section, 12.0 m height (verified: gives 21.6–27s gas residence time depending on final height, see worked cross-check below)
Shell material	FRP (fiberglass-reinforced plastic) or 316L SS, rated for saturated 45–55 °C acidic gas
Mesh Layer 1 (top, coarse)	20 mesh, 0.40 mm wire, 44% open area, Ti Grade 2
Mesh Layer 2 (middle)	80 mesh, 0.13 mm wire, 31% open area, Ti Grade 2
Mesh Layer 3 (bottom, fine)	150 mesh, 0.06 mm wire, 23% open area, Ti Grade 2
Matrix distribution	Trickle-bed, 2.5 ± 0.2 L/s per FIT-301
Pressure drop	$\leq 1,500$ Pa design, SIS trip at 1,800 Pa (Volume 5)
Design code	Pressure vessel code per shell material chosen; FRP per ASME RTP-1 if selected

Tower sizing cross-check (verified correct in original reference):

$$A = 2.5 \times 2.5 = 6.25 \text{ m}^2$$

$$V = 10,000 \times (318/273) = 11,648 \text{ m}^3/\text{hr (at } 45^\circ\text{C)}$$

$$u = 11,648 / (6.25 \times 3600) = 0.518 \text{ m/s}$$

$$H_{\text{required}} = u \times 27\text{s} = 14.0 \text{ m}$$

Specify 12 m with a reduced 21.6 s residence time, or 14–16 m if the Hatta-number finding above requires derating the design capture efficiency and compensating with additional residence time.

6.3 Ultrasonic Demolding System (US-301/2/3) — Corrected Specification**Titanium Property Correction Applied**

Original reference used $\rho = 2,710 \text{ kg/m}^3$ (aluminum's density, mislabeled titanium) and $c = 3,700 \text{ m/s}$, giving a required amplitude of $0.7 \text{ }\mu\text{m}$. Using correct Ti Grade 2 properties ($\rho = 4,510 \text{ kg/m}^3$, $c = 6,070 \text{ m/s}$, the real handbook longitudinal sound speed):

$$A_{\text{req}} = \frac{\tau_{\text{shear}}}{2\pi f \rho c} = \frac{2.2 \times 10^6}{2\pi \times 4.0 \times 10^4 \times 4510 \times 6070} = 0.32 \text{ }\mu\text{m}$$

Table 11: US-301/2/3 Ultrasonic Transducers — vendor RFQ datasheet (corrected).

Parameter	Specification
Type	Langevin-type piezoelectric transducer, Ti Grade 2 horn
Frequency	40 kHz
Required minimum amplitude	0.32 μm (corrected from 0.7 μm ; see box above)
Specified drive amplitude	3.5 μm (practical Langevin transducer output) $\Rightarrow$ safety margin now 10.9 $\times$, not the originally claimed 5 $\times$ — request vendor confirm this is not over-specified for the application (excess amplitude can accelerate mesh fatigue)
Quantity	One per mesh tier (3 total) minimum; vendor to propose array count for uniform coverage across 2.5m $\times$ 2.5m cross-section
Duty cycle	Continuous, with programmable pulse mode for fouling-dependent operation
Acoustic exposure	Design enclosure for $\leq 85 \text{ dBA}$ at 1m operator position per Volume 5 HSE requirements

6.4 CA-KR1 Enzyme Dosing Skid (DS-301) — Corrected Mass Balance

Table 12: DS-301 Enzyme Dosing Skid — vendor RFQ datasheet (corrected duty).

Parameter	Specification
Enzyme	CA-KR1 (thermo-alkaliphilic carbonic anhydrase, <i>Bacillus halo-durans</i> -derived)
Target concentration	12 mg/L in matrix flow (unchanged — this is a concentration, not a mass-flow figure, so unaffected by the CO ₂ correction)
Matrix flow rate	2.5 ± 0.2 L/s (unaffected)
Corrected CO ₂ duty	2,611.5 kg/hr capture target (down from stated 12,255 kg/hr)
Storage	Lyophilized, –20 °C, 24-month shelf life; on-site cold storage for minimum 90-day supply
Dosing pump	Metering pump, 10–15 L/hr range, 316L SS wetted parts
Activity monitoring	CIT-301 online carbonic anhydrase activity monitor, target ≥ 8,500 WAU/mg, alarm below threshold
Pilot verification requirement	Given the Hatta-number finding (Section 4.1), the vendor commissioning protocol (Volume 4) must include a capture-efficiency verification run at design enzyme loading before this skid's dosing rate is finalized for continuous operation.

7 Stage 4 Equipment Specifications — Consolidation and Press Foundry

7.1 Design Duty Summary (Largely Unchanged — See Section 1.4)

Confirmed Duty

Corrected total solids to Stage 4: 6,028.7 kg/hr (vs. original reference's 6,105 kg/hr — within 1.2%). **Original Stage 4 equipment specifications are retained with only the below verification notes.**

7.2 Pug-Mill (PM-401), Hydrocyclone (HC-401)

Table 13: PM-401 / HC-401 — vendor RFQ datasheet.

Parameter	Specification
PM-401 type	Twin-shaft pug-mill, Hardox 400 blades
PM-401 capacity	1.5 m ³ working volume, 45 RPM, 15-min retention (sized for 6,029 kg/hr solids + ash slurry co-feed)
HC-401 type	Polyurethane-lined hydrocyclone, 75 mm apex
HC-401 duty	$\Delta P = 1.5$ bar, 65% underflow solids target
Overflow handling	Recycle to Stage 1 sump per original P&ID

7.3 Hydraulic Press (HP-401) — Verified Force Balance

Table 14: HP-401 Hydraulic Press — vendor RFQ datasheet.

Parameter	Specification
Type	4-cavity rotary press, Hardox 400 dies
Force rating	250 tonnes (verified: 4 cavities $\times$ 51.6 t/cavity = 206.3 t required $\Rightarrow$ 21% margin, correct as stated)
Compaction pressure	200 bar, 30s dwell per cycle
Block dimensions	230 $\times$ 110 $\times$ 75 mm, 3.0 kg finished mass (density check: 1,581 kg/m ³ , verified correct, within IS-2185 band of 1,500–1,800 kg/m ³)
Throughput	2,010 blocks/hr at corrected solids duty (vs. 2,035 claimed — within tolerance, no re-rating needed)
Hydraulic safety	Nitrogen pre-charge accumulator at 175 bar, dual block-and-bleed valve, relief at 220 bar, Category 3 PLd guarding per ISO 13849 (full detail in Volume 5)

7.4 Curing Tunnel (CT-401)

Table 15: CT-401 Curing Tunnel — vendor RFQ datasheet.

Parameter	Specification
Length	18 m, insulated
Operating conditions	55–60 °C, 90% RH, 12 hr residency
Arrhenius rate correction	Original reference's pozzolanic rate constant calculation contained an arithmetic error (stated $k = 0.013 \text{ hr}^{-1}$; correct value from the same Arrhenius parameters is $k = 0.082 \text{ hr}^{-1}$, a $6.3\times$ difference). A higher true rate constant is favorable — it suggests curing may reach adequate C-S-H gel formation faster than the original design assumed. Recommend the commissioning team (Volume 4) run a cure-time sensitivity trial (8h/10h/12h) rather than assuming the original 12-hour figure is the minimum required.
Construction	Insulated steel shell, chemically-resistant PVC or FRP liner

8 Master Bill of Materials

Table 16: Consolidated Bill of Materials — major tagged equipment (Rev. 1, corrected basis).

Tag	Item	Qty	Standard/Code	Notes
D-101	Inlet ductwork, Ø600mm	1 lot	ASME B31.1 or local equivalent	Al ₂ O ₃ -tile lined
V-101	Venturi quench	1	ASME VIII Div. 1	Ceramic throat liner
T-101	Quench sump, 3,500L	1	ASME VIII Div. 1 / API 650	UNS S32750
F-101	ID fan, 35 kW	1	AMCA-rated	VFD included
P-101	Sump pump	1	API 610 or ANSI B73.1	Super Duplex
R-201	Leaching reactor, 15,000L	1	ASME VIII Div. 1, DIN 28136	Glass-lined
T-202	Dissolution tank, 12,000L	1	ASME VIII Div. 1	316L SS + SI trim loop
MX-203	Static mixer	1	Vendor standard	CPVC Sch. 80
P-202	Recirc. pump, 30 m ³ /hr	1	ANSI B73.1	Hastelloy C-276
C-301	Mesh absorber tower	1	FRP per ASME RTP-1 or 316L	3-tier Ti mesh
US-301/2/3	Ultrasonic transducers	3+	Vendor standard	Corrected amplitude spec
DS-301	Enzyme dosing skid	1	Vendor standard	316L SS
PM-401	Pug-mill, 1.5 m ³	1	Vendor standard	Hardox 400
HC-401	Hydrocyclone	1	Vendor standard	PU-lined
HP-401	Hydraulic press, 250t	1	ISO 13849 (safety)	Hardox 400 dies
CT-401	Curing tunnel, 18m	1	Vendor standard	Insulated, FRP/PVC lined

8.1 Consumables and Chemical Inputs (Annual, Corrected Basis)

Table 17: Annual consumable requirements at 7,500 hr/yr, corrected basis.

Item	Rate	Annual Qty	Note
Chitosan (85% DDA)	18.32 kg/hr	137.4 t/yr	Unaffected by correction
BOF slag	250 kg/hr	1,875 t/yr	Unaffected
CA-KR1 enzyme	2.5 mg/hr (conc. basis)	per dosing skid spec	Concentration-based, verify total consumption during pilot
HF (mesh etching)	per Volume 2 procedure	batch, not continuous	Hazardous — see Volume 5
NaBH ₃ CN	per Volume 2 procedure	batch, not continuous	Hazardous — HCN precursor, see Volume 5

9 Procurement Schedule and Long-Lead Items

Table 18: Long-lead procurement items and estimated lead times.

Item	Est. Lead Time	Notes
Titanium Grade 2 mesh screens (C-301)	16–20 weeks	Custom weave for 20/80/150 mesh gradient; qualify vendor early
Glass-lined reactor R-201	20–24 weeks	Glass-lining is a specialty process; long lead is typical
Super Duplex sump T-101	14–18 weeks	Material certification (PREN test) adds time
Hydraulic press HP-401	18–22 weeks	Includes Hardox die fabrication and factory acceptance test
Ultrasonic transducers US-301/2/3	10–14 weeks	Confirm corrected amplitude spec before order placement
Hastelloy C-276 pump P-202	12–16 weeks	Exotic alloy casting lead time
CA-KR1 enzyme (first fill + 90-day buffer stock)	8–12 weeks	Confirm cold-chain logistics with supplier

Procurement Sequencing Recommendation

Place orders for Titanium mesh, glass-lined reactor, and Super Duplex sump **before** finalizing the Stage 3 pilot verification (Section 4.1). If the Hatta-number/enhancement-factor open item requires a larger enzyme loading or longer residence time to hit the 95% capture target, the mesh tower height (and therefore the Ti mesh screen quantity) could change. Recommend a design freeze gate at the end of Volume 4's pilot test before releasing the Ti mesh purchase order to avoid costly rework.

10 Vendor Qualification Criteria

Table 19: Minimum vendor qualification criteria by equipment category.

Category	Minimum Qualification
Pressure vessels (V-101, T-101, R-201, T-202)	ASME U-stamp or equivalent national certification; minimum 5 years' experience with the specified material of construction; provide 3 reference installations
Titanium mesh fabrication	Demonstrated experience weaving Ti Grade 2 wire below 0.10mm; provide mill certs for wire source
Hydraulic press	ISO 13849 safety-category compliance; factory acceptance test (FAT) witnessed by purchaser
Enzyme supply (CA-KR1)	GMP or equivalent quality system; certificate of analysis per batch (WAU/mg activity, purity by SDS-PAGE)
Glass-lining specialists	DIN 28136 or ASME-equivalent certification; thermal shock test data

11 Site Civil and Structural Requirements

Table 20: Site footprint and structural design loads (per original reference, verified consistent).

Area	Requirement
Stage 1 stack + sump	50 m ² , 9m clear height, foundation rated for T-101 full+liquid weight plus seismic factor per local code
Stage 2 reactor + tank farm	80 m ² , secondary containment bund sized for 110% of largest vessel volume
Stage 3 mesh tower	40 m ² , 12m clear height, wind load design per local code (tall slender structure)
Stage 4 press + curing tunnel	120 m ² , reinforced floor slab rated for 250t press dynamic loading plus impact factor
Chitosan + slag storage	60 m ² , covered, slag storage segregated from chitosan (dust/-moisture control)
DCS room, lab, HSE	50 m ² , positive-pressure HVAC if located near HF/HCN handling areas (Volume 5)

A Appendix A: Blank Vendor Datasheet Template

Field	Vendor to Complete
Equipment tag	
Service description	
Design flow / duty	
Design pressure / temperature	
Materials of construction (wetted)	
Corrosion allowance	
Design code	
Inspection & test plan reference	
Weight (dry / operating)	
Nameplate data required	
Warranty period	

Table 21: Blank equipment datasheet template for vendor RFQ issuance.

B Appendix B: Inspection and Test Plan (ITP) Template

Inspection Item	Method	Acceptance Criteria	Witness Level
Material certification	Mill test certificate review	Matches spec MOC	Document review
Dimensional check	Physical measurement	Per drawing $\pm$ tolerance	Hold point
Weld inspection	RT/UT/PT per code	Zero defects per code	Hold point
Pressure test	Hydrostatic per code	1.5 $\times$ design pressure, no leaks	Witness
Liner adhesion (if applicable)	Vendor standard pull test	Vendor spec	Document review
Performance test (rotating equip.)	Factory test per API/ANSI	Within 5% of rated duty	Witness

Table 22: Generic Inspection and Test Plan (ITP) template — customize per equipment category.

C Appendix C: Cross-Reference to Verification Report Findings

Verification Finding	Where Corrected in This Volume	Downstream Volume Impact
CO ₂ mass flow 4.7 $\times$ inflated	Section 1.2, 1.3	Volume 4 (performance test basis), Volume 6 (SOPs reference correct tonnage)
Calvert d ₅₀ inconsistent with inputs	Section 3.2	Volume 4 (empirical PM10 verification required)
Stage 1 ΔP off by 10 $\times$	Section 3.2	Volume 5 (SIS setpoint unaffected, still 5 kPa)
Saturation index sign error	Section 4.2 (Stage 2)	Volume 5 (HAZOP: scaling risk in T-202)
Hatta number off by 10 $\times$, enhancement factor unsupported	Section 5.1, 5.4	Volume 4: mandatory pilot verification before full-scale procurement
Titanium density/sound-speed inconsistency	Section 5.3	None (corrected spec is more conservative amplitude)
Arrhenius rate constant off by 6.3 $\times$	Section 6.3	Volume 4 (cure-time sensitivity trial recommended)
Exergy destruction unit error	Not equipment-relevant	Volume 4 (informational only, no hardware impact)
Revenue total inconsistent with line items	Not applicable to hardware	Financial planning only, out of scope for this volume